Andrew W. **Lyjak**

Software Development and Verification · Systems Engineer

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"This is the real secret of life – to be completely engaged with what you are doing in the here and now. And instead of calling it work, realize it is play."

November 15, 2024

To the Latitude AI Recruitment Team

Early in my career I realized that the context and affordances accessible to an individual can dramatically impact their decision making capability. Cultivating tools and environments that improve quality of thought has been my calling ever since. Therefore, Systems Engineering is my natural home, and I excel at creating environments where systems thinking is spread throughout the organization rather than extracted into it's own domain of expertise. Although more difficult to implement, a culture where systems thinking is integrated across the organization results in higher quality results than if systems analysts were wholly responsible for controlling verification, requirements, and interfaces throughout the project. It results in empowered and competent teams who have the skills and tools at their disposal to solve and communicate complex problems wherever those problems exist – be it within or between the analytical, manufacturing, and regulatory specialties responsible for making a complex multi-dimensional engineering effort succeed or fail.

At SpaceX I learned that it is more important to plan on continuous improvement from the onset than to plan for perfection. The fastest way to improve, both personally and product-wise, is to keep executing. Post-SpaceX, I have used my experience to design and implement engineering coordination systems, including a continuous integration platform for Raytheon, and a production workflow platform for my last salaried employer, H3D. Presently, I am developing a Production Lifecycle Management application that integrates digital procedures with automation and digital sensing.

I have expertise in a wide range of topics, including "hard science" fields such as aerospace, electrical, and computer engineering, as well as "soft science" fields such as Requirements, Verification, Certification, Safety, Quality, and Reliability analyses. One noteworthy effort I lead at SpaceX was the entire CBCS analysis and verification plan for the Crew and Cargo variants of the Dragon spacecraft. As part of this effort I and my collaborators developed and executed a Fault Tolerance Analysis specialized in accounting for the networked nature of Computer Based Control Systems – systems where failure triggers and responses cannot be assumed to be triggered from deterministic sources.

In closing, my background, skills, and motivations are well suited to the Senior Systems Engineer, Fault Handling Systems, as well as Staff Functional Safety Engineer, Runtime Software & OS positions. I, and my family, are enthusiastic about the opportunity to join you. Thank you for reviewing my application and I hope to hear from you soon.

Sincerely,

Andrew W. Lyjak